Prediction of drug combinations based on efficacy and safety estimates

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Abstract
Identification of drug combinations to treat complex diseases like cancer necessitates a balance between efficacy and safety. Lately, a combination of network biology and machine learning has been successful due to their ability to put in a single framework the different entities related to drugs and diseases and discover hidden patterns in vast datasets. Previous research has shown that network distances between proteins effected by diseases and those targeted by drugs could be effectively used for identifying drug combinations. However, most current methods address the aspects of efficacy and safety separately. Here we present a method that uses network derived efficacy and safety estimates to train supervised machine learning classifier for predicting effective and safe drug combinations. Initial assessment of the method using leukemia and lung cancer drug combinations showed promising results for random forest-based models which achieved test accuracy greater than 0.9 and 0.7, respectively. Currently, efforts are underway to validate the models using external dataset and to improve the feature engineering used to create the features for assessment of safety and efficacy. The developed method can aid in pre-assessment of efficacy and safety of drug combination before entering clinical trials and thus reducing the number of failures.

Introduction
Drug combination therapy have emerged as a promising solution to treat complex diseases like cancer due to its ability to target multiple pathways simultaneously. Apart from this they can also help avoid toxicity by reducing the required doses for the individual drugs. However, not all combinations may be effective and safe at the same time. Current in silico methods for identification of drug combinations tend to over emphasize synergism or effectiveness and overlook the safety aspect. Also, more often they do not predict disease specific drug combinations. This calls for a need to device efficient computational methods that can prioritize disease specific drug combinations based on both efficacy and safety simultaneously.

Method
Our approach leverages on using network derived features related to efficacy and safety to train supervised machine learning classifier for distinguishing effective and adverse drug combinations. A critical step underlying this approach is designing efficacy and safety estimates. For addressing efficacy, we build libraries of known disease related genes of different types as those based on genetic association, differential expression, literature mining, etc. from databases like OpenTargets, DisGeNet and PharmGKB, and from tools like TheTA. On the other hand, for estimating adverse reactions, we build gene libraries of adverse drug reactions (ADRs) reported to be common to the drugs for the disease based on information in ADReCS. A combined library was also prepared consisting of disease related gene sets and ADRs that have been reported to be common for drug withdrawal (hepatotoxicity, immune-related reaction, cardiotoxicity, neurotoxicity, haematological toxicity and carcinogenicity). These gene libraries are used to understand how well the genes perturbed by the drug combinations, which in turn is identified using random walk with restart (RWR) on a STRINGdb retrieved protein-protein interaction (PPI) network starting from the known drug targets as seeds, enrich efficacy and safety related genes. The estimates or normalized enrichment scores for the top 500 genes with highest probability obtained from RWR were finally used to train three machine learning models – glmnet, random forest (RF) and support vector machine (SVM) – for classification of the drug combinations.

Results
The method was tested using drug combinations for lung cancer and leukemia. In absence of gold standard drug combination datasets, we retrieved the drug combinations based on reported synergistic effect and ADRs in DrugCombDb and DrugBank. We observed that RF models performed better in both cases. The use of SMOTE had significant impact in models for lung cancer which had imbalanced classes. Analysis of the features selected by the classifiers revealed that gene sets retrieved from PharmGKB were more relevant especially for classifiers trained using efficacy or combined features. The gene sets from OpenTargets (literature) and DisGeNet (curated) also occurred among the important features for the classifiers. Further, for the combined library based classifier, efficacy related features were more frequently selected. However, for classifiers trained using safety related features, there was no consensus in the selected features even though the classifiers attained high accuracy. This might be due to the inclusion of large number of features for training. We compared our model trained for lung cancer with PRODeepSyn, a deep learning based tool that utilizes PPI and omics data to predict synergistic drugs. We observed that drug combinations predicted as effective by our tool were also predicted as synergistic by PRODeepSyn in lung cancer cell line SKMES1.

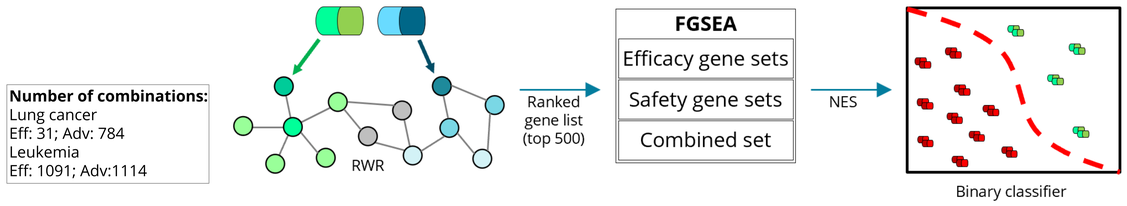

Figure: Overview of the pipeline for training classifier for distinguishing effective and safe drug combinations

Table: Average area under curve & standard deviations of the models

		Efficacy Library		Safety Library		Combined Library	
Disease	**Model**	**noSMOTE**	**SMOTE**	**noSMOTE.**	**SMOTE**	**noSMOTE**	**SMOTE.**
Leukemia	**rf**	0.982 (0.006)	0.984 (0.006)	0.997 (0.004)	0.995 (0.003)	0.984 (0.008)	0.986 (0.006)
	glmnet	0.818 (0.02)	0.809 (0.011)	0.992 (0.003)	0.991 (0.005)	0.82 (0.018)	0.825 (0.014)
	svmRadial	0.984 (0.004)	0.98 (0.01)	0.985 (0.019)	0.966 (0.032)	0.987 (0.007)	0.983 (0.007)
Lung Cancer	**rf**	0.55 (0.07)	0.827 (0.04)	0.558 (0.04)	0.902 (0.073)	0.558 (0.069)	0.782 (0.08)
	glmnet	0.516 (0.034)	0.755 (0.105)	0.733 (0.077)	0.879 (0.05)	0.541 (0.044)	0.728 (0.043)
	svmRadial	0.625 (0.059)	0.677 (0.098)	0.591 (0.062)	0.728 (0.088)	0.574 (0.048)	0.72 (0.095)

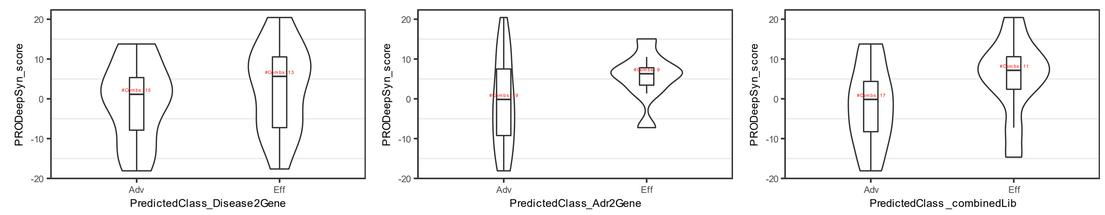

Figure: Distribution of PRODeepSyn synergy scores for 28 drug combinations predicted as adverse and effective using model trained for lung cancer

Conclusions
Initial comparative assessment of our method with PRODeepSyn shows that our method may be better at predicting drug combinations that are effective as well as safe. While drug combinations predicted as effective by our method also had higher synergy score from PRODeepSyn, some combinations which were predicted as adverse were also predicted as synergistic by PRODeepSyn.

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